D. Y. PATIL COLLEGE OF ENGINEERING AND TECHNOLOGY, KOLHAPUR
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING



A
Project- II Report on

"Patient Health Information Exchange Using Blockchain"

Submitted by:

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Group Number: G5 **Class: B.Tech (CSE)** **Division: A**

Year:2022-2023 Sem-II

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(AN AUTONOMOUS INSTITUTE)
KASABA BAWADA, KOLHAPUR

CERTIFICATE

This is to certify that the project group consisting of following members have satisfactorily completed theproject-II work entitled "Patient Health Information Exchange Using Blockchain " at BTech (CSE) semester – VIII as prescribed in the syllabus of Shivaji University for the academic year 2022-2023.

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INTRODUCTION

PROJECT INTRODUCTION

The electronic health record (EHR) could be a digital record that firmly records data a few patients and makes it accessible to users in period of time.

Health data Exchange refers to the method of act patients' aid statistics between aid patientsand aid professionals. There has been attention on health-statistics exchange as a major toolfor providing wonderful and economical treatment, and several other policy approaches are accustomed strengthen competencies during this space. Health data exchange (HIE) delivers vital advantages to patients by up aid quality and expediting coordinated treatment.HIE's have evolved into 3 differing types, every providing a novel approach to health data access. Direct Exchange: Health care professionals will safely share patient information with alternative physicians through a directed exchange. The info is transmitted over the web in a very dependable and secure encrypted manner. Data privacy is ensured. Directed exchange is employed to transmit quality-measurement results to the agency and vaccination data to public health agencies. Query-based Exchange: Through the query- based exchange, physicians could ask for and retrieve precise health data on a patient from whichever acceptable clinical sources are accessible. This kind of rush is most usually utilized by suppliers to convey unlooked-for treatment, like in associate degree emergency circumstance or once a patient goes into labor whereas move. Consumer-Mediated Exchange: Patients will access their own medical data firmly and simply on-line. This enables for the patient to participate in their own care and will increase satisfaction and accuracy. Through this utilization of rush, customers will higher manage their own health care, not like those who could use on-line banking to manage their finances.

Electronic Health Information Exchange is a system that allows doctors, nurses, health care providers, pharmacists, and patients to access and share the patient’s medical information. This sharing mechanism undergoes greater efficiency and speed, high quality, cost of patientcare, and safety.

Despite the conventional methods of storing the data on paper, the electronic health care information helps share the communication between the providers through mails, fax, or patients themselves, without any need to carry the information from appointment to appointment.

NEED OF WORK

- **Seamless Care Coordination**: PHIE enables the secure exchange of patient health information among healthcare providers, ensuring that crucial data is available when and where it is needed. This improves care coordination, especially in cases where patients receive treatment from multiple providers or transition between different healthcare settings. Instant data delivery.

- **Enhanced Patient Safety and Quality of Care**: Access to comprehensive and up-to-date patient health information through PHIE reduces the risk of medical errors, such as adverse drug interactions or duplicate tests. Healthcare providers can make more informed decisions, leading to improved patient safety and better healthcare outcomes.

- **Efficient Healthcare Delivery**: PHIE streamlines the exchange of health information, eliminating the need for manual processes like faxing or mailing records. It reduces administrative burden, enhances workflow efficiency, and saves time for healthcare professionals, allowing them to focus more on patient care.'

- **Timely Access to Critical Information**: In emergency situations or urgent care scenarios, quick access to patient health information can be life-saving. PHIE ensures that essential data, such as allergies, medications, and medical history, is readily available to healthcare providers, enabling faster and more accurate decision-making.

- **Continuity of Care**: PHIE supports the continuity of care as patients move across different healthcare settings or encounter various providers. It enables the seamless transfer of patient health records, ensuring that the receiving healthcare professionals have access to the necessary information for ongoing treatment and management.

PROBLEM STATEMENT

HIE or the Health Information Exchange can be described as a system that allows the Healthcare sector workers, the doctors, to access and share the important medical information of the patients associated with them. This process of sharing has to undergo a lot of scrutiny for maintaining the quality of it and assuring the safety of patient's information.

OBJECTIVES

- Removes the need for an intermediary as all users have access to a distributed ledger for conducting secure information exchange, irrespective of brokered trust.
- The distributed framework enables the use of public and private digital identifiers. Cryptography makes them more secure and authentic patient digital identifiers.
- Disintermediation reduces the cost factor, and real-time operation makes the system more efficient.
- Data sharing enables real-time updates throughout the network. Therefore, records remain compatible without affecting interoperability.
- The blockchain-based Smart contracts build consistency and regulations-based methods with consent to access the patient information for selected health organizations.

REQUIREMENT ANALYSIS & SPECIFICATIONS:

INFORMATION GATHERING

- IT is clearly defined, objectives, User's requirement, a realistic timeline and ways to fairly and efficient divide work within the time

- Literature survey was carried on by studying these papers:
  [1]https://www.leewayhertz.com/blockchain-health-information- exchange/#
  [2]https://www.healthit.gov/sites/default/files/page/2019-04/AHAEHRUseDataBrief.pdf .

- For this project contacted those people who actually face the problems during lockdown busy schedule people, emergency patients and elder peoples

- By using various technology as a development of the system. The information about this technology can collect from
  1)https://blog.chain.link/how-to-build-a-dapp/

- The information about how to make Dapps more interactive using button and text can collect from
  1)https://www.udemy.com/course/complete-dapp-solidity-react-blockchain-development/

PROJECT SURVEY:

We gathered as many articles as possible related to our domain including primary and secondary articles. This review paper explores how blockchain technology can revolutionize our current healthcare infrastructure. The sector comprises of different healthcare players (providers, payers, patients, vendors, manufacturers, and research institutes, etc.) performing different roles with different needs. This paper attempts to provide a deeper insight by discovering the scope of blockchain for all healthcare players ina single study and to identify the challenges of blockchain technology in the said sector.

Blockchain and healthcare industry:

A recent survey across industries found that 60 percent of blockchain early adopters are experimenting with a radically different business model or business process. This reinvention should include both back-end and consumer-facing processes. Organizations should not limit themselves to one or the other, but rather pursue both. For us in healthcare, the opportunities to improve care management, patient outcome and patient engagementare significant.

LIFE CYCLE MODEL:

WATERFALL LIFE CYCLE MODEL:

The Waterfall model is a linear and sequential approach to software development that consists of distinct phases that are executed in a predetermined order, with each phase dependent on the completion of the previous one. The typical phases in the Waterfall model include:

Requirements Gathering: This phase involves gathering and documenting the specific requirements of the software or system being developed. In the context of a PHIE project, this phase would involve identifying the requirements for the exchange of patient health information, including data formats, security protocols, and integration requirements with different healthcare systems.

System Design: In this phase, the overall system architecture and design are developed based on the gathered requirements. For a PHIE project, system design would involve designing the infrastructure, data models, and interfaces required for the secure and efficient exchange of patient health information.

Implementation: The implementation phase involves coding and developing the software or system based on the design specifications. In the context of a PHIE project, this would involve implementing the necessary software components, interfaces, and security measures required for data exchange and interoperability.

Testing: Once the implementation is complete, the testing phase begins. This phase involves various types of testing, such as unit testing, integration testing, and system testing, to ensure that the developed system functions as intended and meets the specified requirements. In the case of a PHIE project, testing would involve validating the data exchange processes, ensuring data integrity and security, and verifying interoperability with different healthcare systems.

Deployment and Maintenance: After successful testing, the system is deployed and made operational. This phase involves installing the system in the production environment and ensuring its ongoing maintenance and support. For a PHIE project, deployment would involve integrating the exchange platform into the existing healthcare infrastructure, training users, and establishing mechanisms for ongoing system maintenance and updates.

SOFTWARE REQUIREMENT SPECIFICATION (SRS):

INTRODUCTION:

Purpose of this Document:

To create a UI based decentralized application by making electronic health records (EHRs) more accessible, more accurate, more secure, and less expensive to create and maintain.

Scope of this document –

- To improve quality of care by providing more complete patient information to enhance medical decision making.
- To begin to reduce costs through the elimination of unnecessary duplicate tests.
- To support continuity of care as patients transfer among health care facilities, an aspect growing in importance with the aging population.
- To improve the information flow to health departments and facilitated strongermanagement of emerging infectious diseases.
- To offer innovative ways to conduct population-based research through large databases created by some health information exchange organizations (HIO's)

SOFTWARE REQUIREMENTS

Tools

- Windows 7 or above OS
- Visual Studio code (VS Code)
- MetaMask Browser Extension
- Ganache

Hardware Requirements

- Processor: - Intel i5 Core
- RAM: - 4GB or more
- Hard Disk : - 64 GB or more

DESIGN:

SYSTEM ARCHITECTURE:

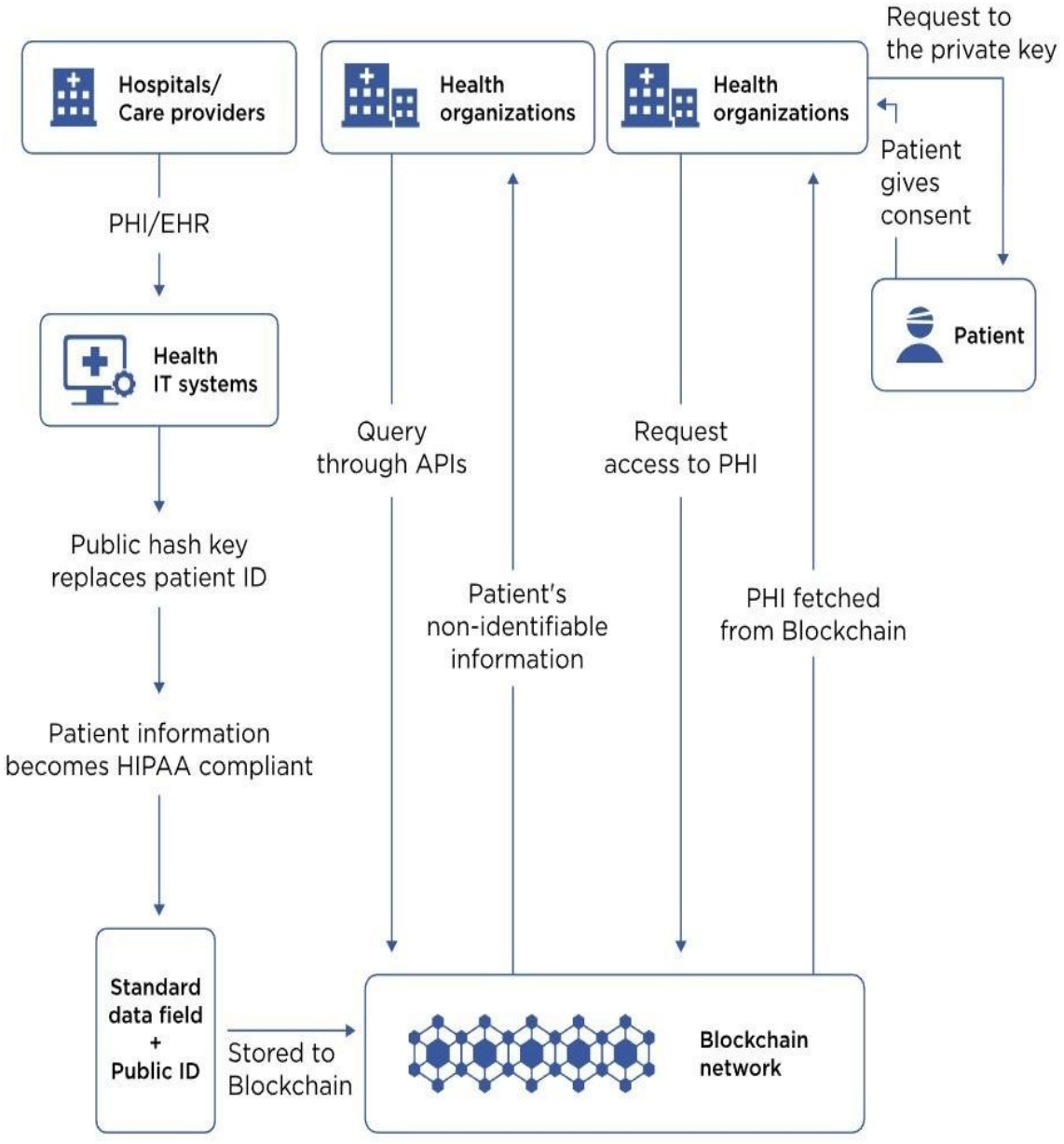


Fig 1: Blockchain work in Health Information Exchange system

1. Make use of the public hash key to replace the patient's identity.

Hash represents a unique numeric value. This hash is used to replace the patient's identity. When a hash replaces a user's identity, attackers cannot decode the identity and the protected health information (PHI). Due to de-identification, the data regarding the patient's name, address, financial information and security number gets removed.

2. Store information on the blockchain and transactions get executed with Unique Identifier

When health organizations or care providers provide service to patients, the patient's entire data gets saved with a public key on the blockchain. The patient's data may include gender, age, date of birth, diagnosis, care plan, treatment procedure, etc. The information provided by health vendors, care providers, or insurance companies gets stored on the blockchain in the form of transactions by triggering smart contracts. If any healthcare provider needs to access a patient's public data, the information gets displayed only after matching the transaction ID.

3. Patient shares the public /private key with Healthcare Units

The public keys of patients are directly linked to the blockchain data. When the patients share the public or private keys with healthcare units, their information becomes visible. When it comes to information storage, blockchain as a transaction layer can store the data in one way:

- **On-chain –** Here, the information is stored directly on the blockchain. For example, hash codes, transactional information, audit records and the metadata about the off-chain storage.

MODULAR STRUCTURE

Admin –

Admin module allows administrators to manage user accounts within the PHIE system. It includes functionalities for creating, updating, and deleting user accounts, as well as assigning roles and permissions to different user types, such as doctors, patients, or other administrators

The admin module focuses on system security and compliance with relevant regulations and standards. It includes features for managing security certificates, configuring encryption, implementing access controls, conducting security assessments

Doctor -

The doctor module incorporates user authentication mechanisms to ensure that only authorized doctors can access patient health information.

The doctor module enables doctors to securely access patient records stored within the PHIE system. It provides a user interface that allows doctors to search for specific patients, view their medical history, diagnoses, treatments, medications, and other relevant health information. The module ensures that doctors can access the necessary patient data based on their role and permissions.

Patient-

The patient module typically includes a patient portal, which serves as the interface for patients to interact with the PHIE system. The patient portal provides a secure and user-friendly environment for patients to access their health records, communicate with healthcare providers, and manage their health-related information. The patient module enables patients to view and manage their personal health records. Patients can access their medical history, allergies, medications, lab results, immunizations, and other relevant health information.

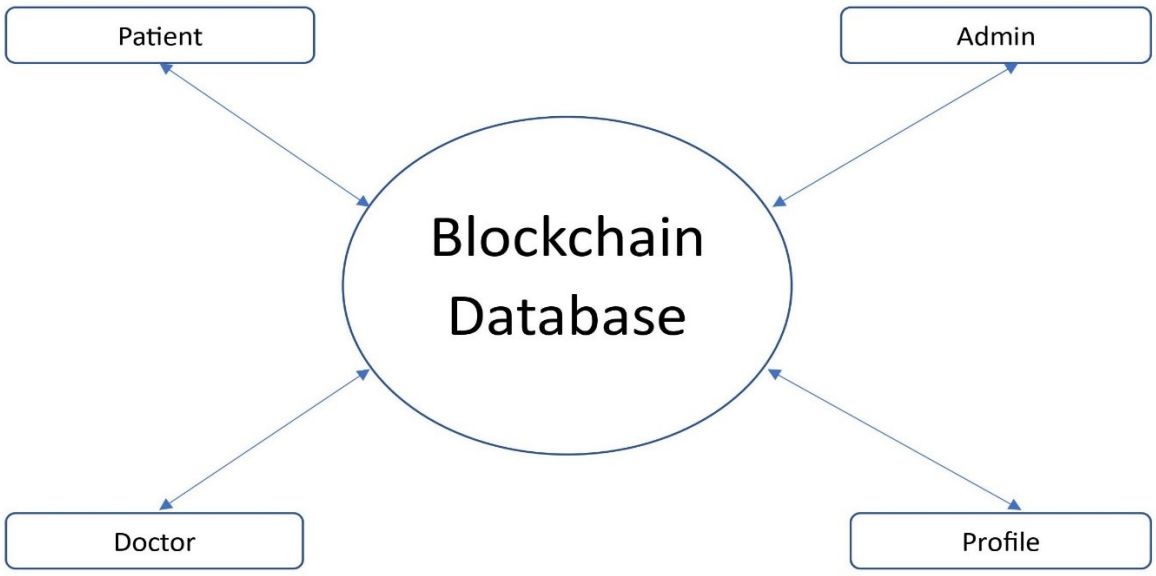
The patient module may include features for patients to schedule appointments with healthcare providers. Patients can check the availability of healthcare professionals, book appointments, reschedule or cancel appointments as needed. The patient module allows patients to update their personal health information, such as contact details. It ensures that patient information remains accurate and up to date, enhancing the quality of care and patient safety.

DATA FLOW DIAGRAMS

This is the zero-level data flow diagram of PHIE, where we have elaborated the high- level process of PHIE system.

High Level Entities and process flow of PHIE system

- Managing all the Doctor
- Managing all the Patient
- Managing the Admin
- Managing all the Profile activities

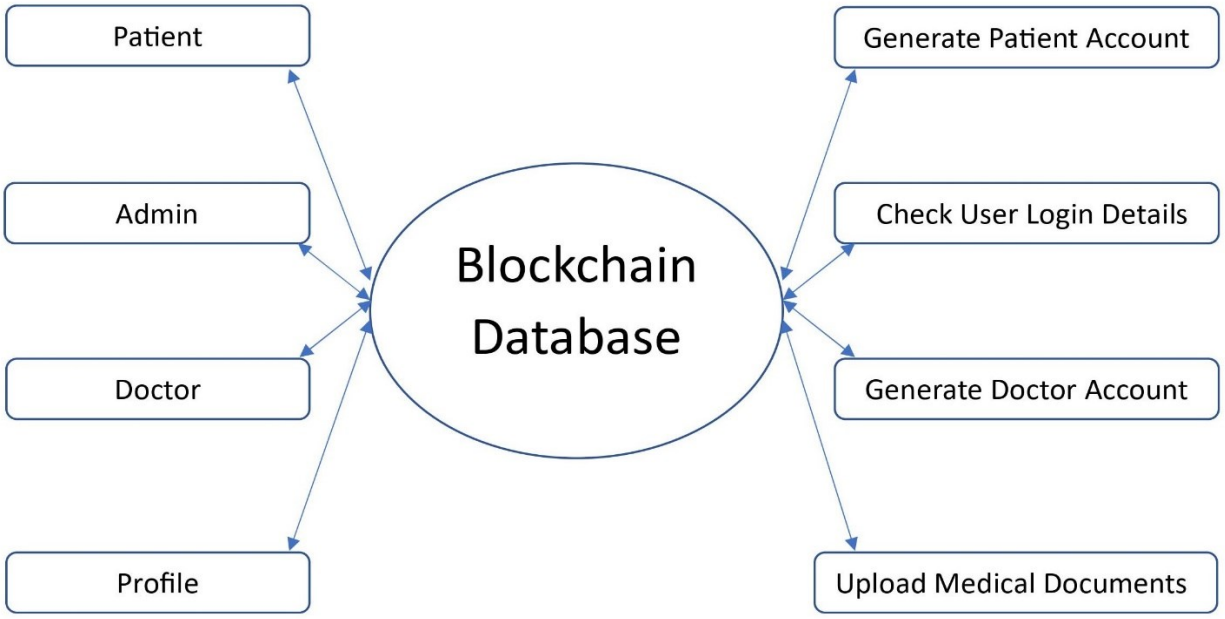


Zero Level DFD – PHIE System

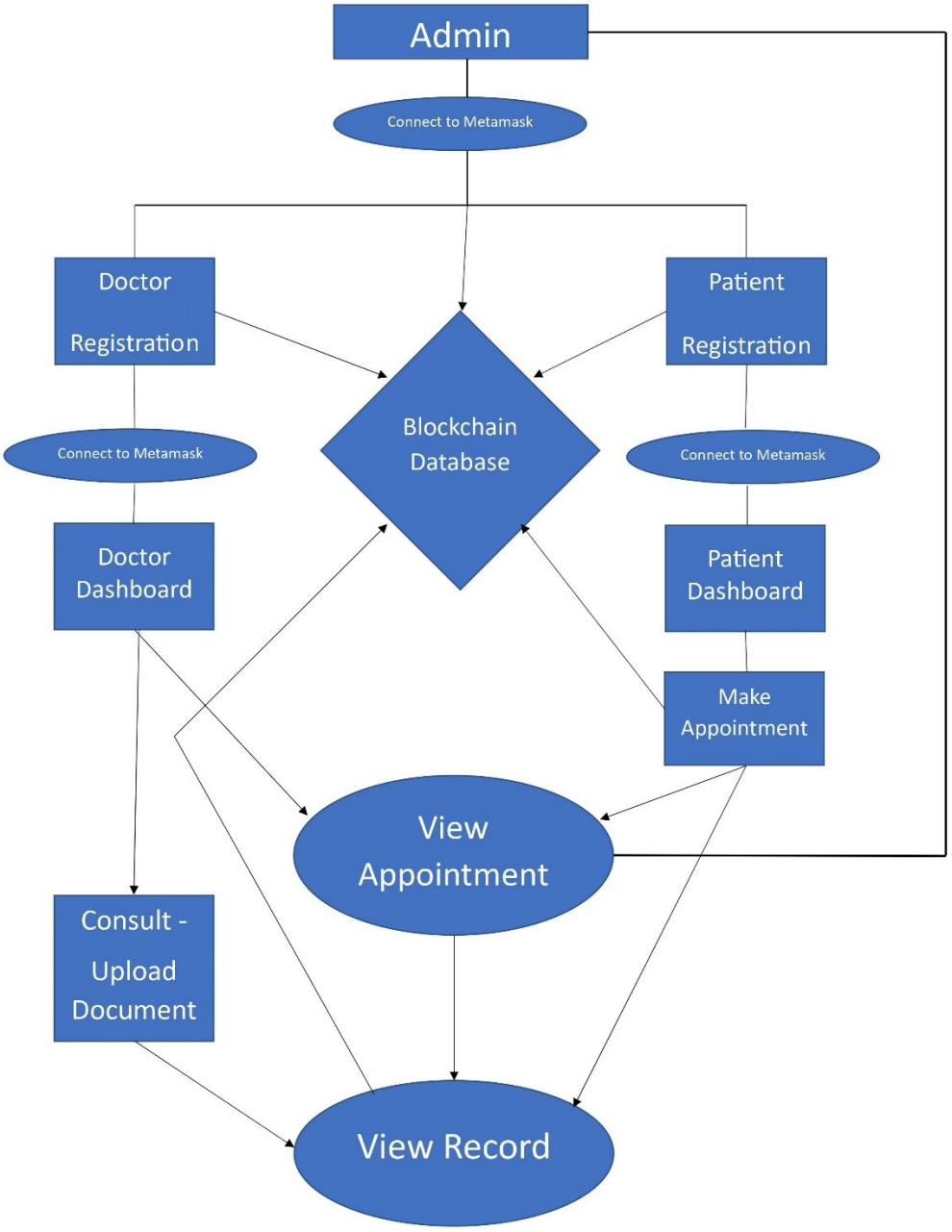
First level data flow diagram of PHIE system: First level data flow diagram of PHIE system shows how the system is divided into sub-systems(processes), each of which deals with one or more of the data flows to or from an external agent, and which together provide all of the functionality of PHIE system as a whole.

Main entities and output of First Level DFD:

- Processing Doctor records and generate Doctor Account
- Processing Patient records and generate Patient Account
- Processing Admin records and check user login details
- Processing Profile records
- Processing Documents records and upload medical documents

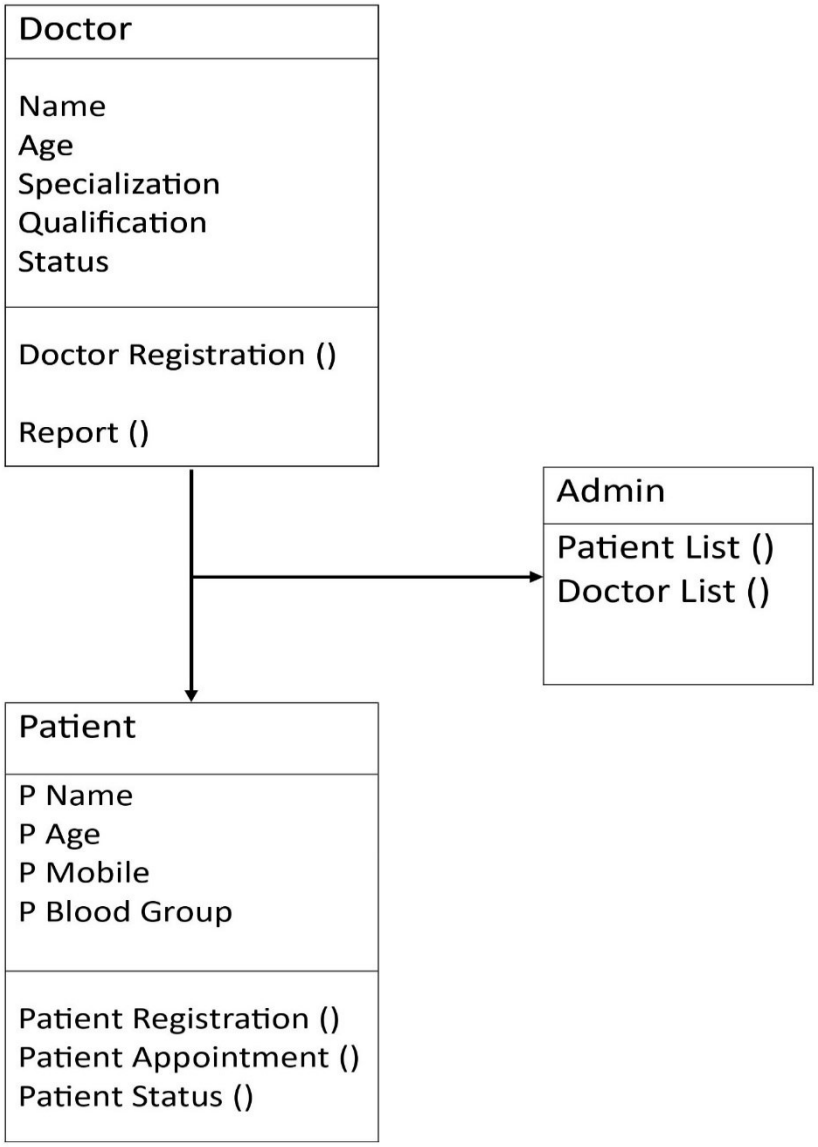


First Level DFD – PHIE System



Second Level DFD – PHIE System

Class Diagram:



TECHNOLOGY USED:

Blockchain

Blockchain technology may be a framework that maintains public group action information, conjointly known as blocks, in various databases, mentioned as the "chain," inan extremely large network connected by peer-to-peer nodes. Blockchain, also known as Distributed Ledger Technology (DLT), employs decentralization and cryptographic hashing to create an immutable and visual history of any digital asset. Kind of storage is usually mentioned as a "digital ledger."

Smart Contract

Smart contracts are typically written in programming languages specifically designed for blockchain platforms. Solidity is the most common language used for writing smart contracts on the Ethereum platform, while other platforms have their own languages .smart contracts are immutable, meaning they cannot be modified or tampered with. The execution and results of smart contracts are recorded on the blockchain, creating a permanent and auditable record of all transactions and actions performed.

SQLite

SQLite is designed as an embedded database, which means it is integrated into the application itself rather than running as a separate server process. This allows applications to have a self-contained database system without the need for a separate database server installation SQLite requires minimal configuration and administration. There is no need for separate user management, access control, or complex setup processes. Applications can simply access and utilize the SQLite database file directly.

JSON

JSON is widely used for data exchange between web services and APIs. It is commonly used in web applications to send data from the server to the client and vice versa. JSON is also used as a configuration file format and for storing structured data in databases.

IPFS

IPFS aims to create a global, distributed file system that allows users to address files based on their content rather than their location. It breaks files into small chunks and assigns unique cryptographic hashes to each chunk. These hashes serve as addresses, enabling efficient storage and retrieval of files across the network.

IPFS uses content addressing, meaning that files are identified and accessed based on their content rather than by their location or assigned names. This approach provides benefits such as deduplication, integrity verification, and the ability to access files even if their original source is offline.

Web3

Web3 technology, also known as the decentralized web or Web3, refers to a set of technologies and principles that aim to enable decentralized, peer-to-peer, and trustless interactions on the internet. While Web3 technology is not specifically tied to patient health information exchange (PHIE), it can have implications for the development and implementation of PHIE systems. Web3 technology leverages blockchain and decentralized protocols to provide enhanced privacy, security, and data ownership.

Language Used:

Angular is a platform and framework for building single-page client applications using HTML and TypeScript. Angular is written in TypeScript. It implements core and optional functionality as a set of TypeScript libraries that you import into your applications.
Frameworks in general boost web development efficiency and performance by providing a consistent structure so that developers don't have to keep rebuilding code from scratch. Frameworks are time savers that offer developers a host of extra features that can be added to software without requiring extra effort.

Python's simplicity and ease of use make it well-suited for automating repetitive tasks and scripting. In the context of PHIE. Frameworks like Django provide a robust foundation for developing web-based PHIE solutions, allowing developers to build secure, scalable, and user-friendly interfaces for accessing and managing patient health information. In the context of PHIE, Python can be used to process and analyze health-related data, such as patient records, medical imaging data, lab results, and clinical trial data.

HTML (Hypertext Markup Language) is the standard markup language used for creating and structuring web pages and applications. It provides a set of tags and elements that define the structure, content, and presentation of web documents.

SASS (SASS CSS) is a CSS preprocessor that extends the functionality of CSS by adding features like variables, nesting, mixing, and more. in the context of PHIE, SASS can be utilized to enhance the styling and organization of the user interface (UI) components of the application. By using SASS, you can define reusable styles, create nested selectors for easier readability, and leverage variables for consistent theming across the application.

TOOLS:

MetaMask acts as a digital wallet that allows users to securely store and manage their Ethereum-based cryptocurrencies. In the context of PHIE, if the system utilizes a blockchain that operates on Ethereum or requires cryptocurrency transactions, users can utilize MetaMask to hold the necessary tokens and perform transactions securely.

User Authentication: MetaMask provides a convenient and secure method for user authentication and authorization. PHIE systems can leverage MetaMask's authentication capabilities, allowing users to sign transactions or provide consent for accessing their health information securely and with the control of their private keys.

Ganache is a personal, local Ethereum blockchain developed by Truffle Suite. It is commonly used for local development, testing, and debugging of Ethereum smart contracts and decentralized applications (DApps).Ganache provides a local blockchain environment that runs on your own machine. It allows developers to create a private Ethereum network for testing and development purposes without the need to connect to the public Ethereum network or deploy contracts to a live network.

ALGORITHM

- The Algorithm explains the functioning of the smart contract for patient records. This algorithm has five functions that are to define roles, add, view, update and delete records
- **Assign Roles :**

function Define Roles (New Role, New Account)
add new role and account in
roles mapping
end function

- **Add Data :**

function Add Patient Record (*contains variables to add data*)
if (Msg.Sender == doctor) **then**
add data to particular patient's record
else Abort session
end if
end function

- **Retrieve Data :**

function View Patient Record (patient id)
if (msg.sender == doctor || patient) **then**
if (patient id) == true **then**
retrieve data from specified patient (id)
return (patient record)
to the account that requested the retrieve operation
else Abort session
end if
end if
end function

- **Update Data :**

```
function Update Patient Record ( contains variables to
update data)
    if ( msg.sender == doctor ) then
        if( id == patient id && name == patient name ) then
            update data to particular patient's record
        return success
        else return fail
        end if
    else Abort session
    end if
end function
```

- **Delete Data:**

```
function Delete Patient Record ( patient id )
    if (msg.sender == doctor ) then
        if ( id == patient id ) then
            delete particular patient's record
            return success
        else return fail
        end if
    else Abort session
    end if
end function
```

IMPLEMENTATION AND CODING

As visible in below figures the proposed framework or system has three modules. These modules when combined together would keep our system working. These modules have further concepts that need to be understood they are explained as follows.

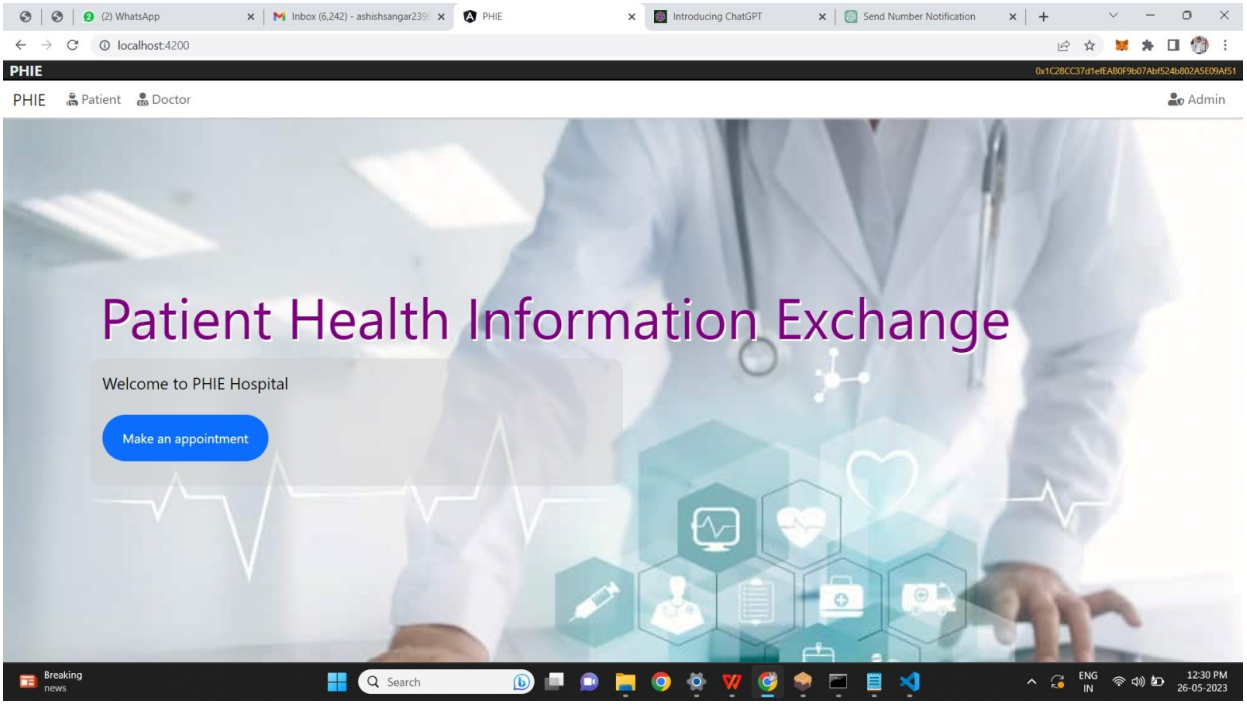


Fig: Home Page

The admin function can only be accessed by System Administrator .If Any others try to Access the function it throws an Error.!! Message.

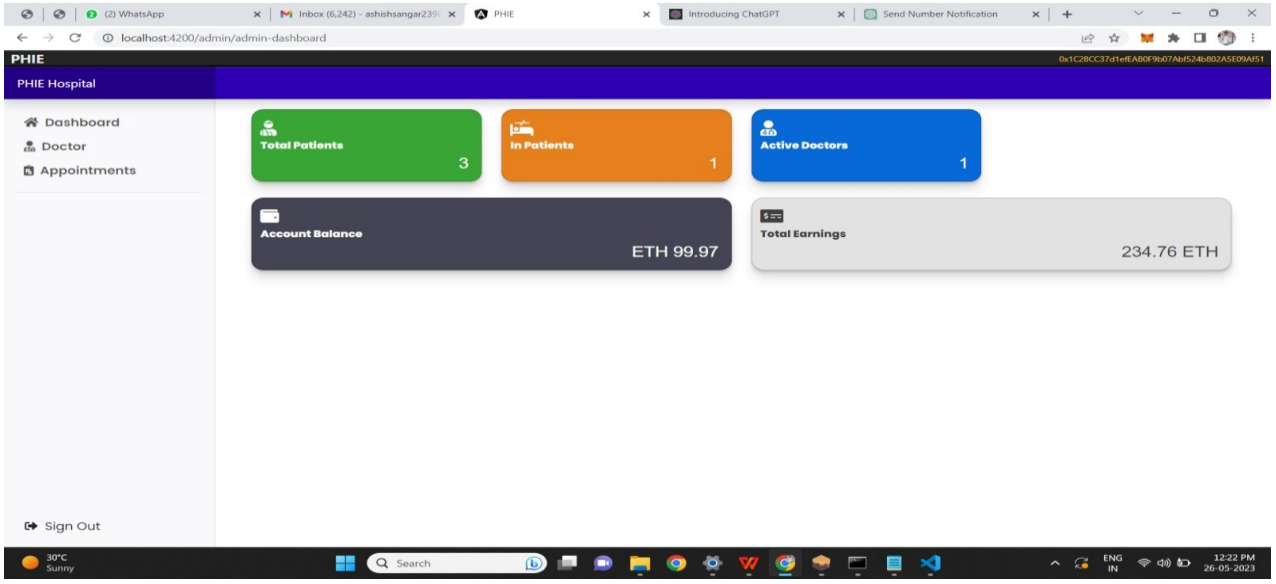


Fig. Admin Dashboard

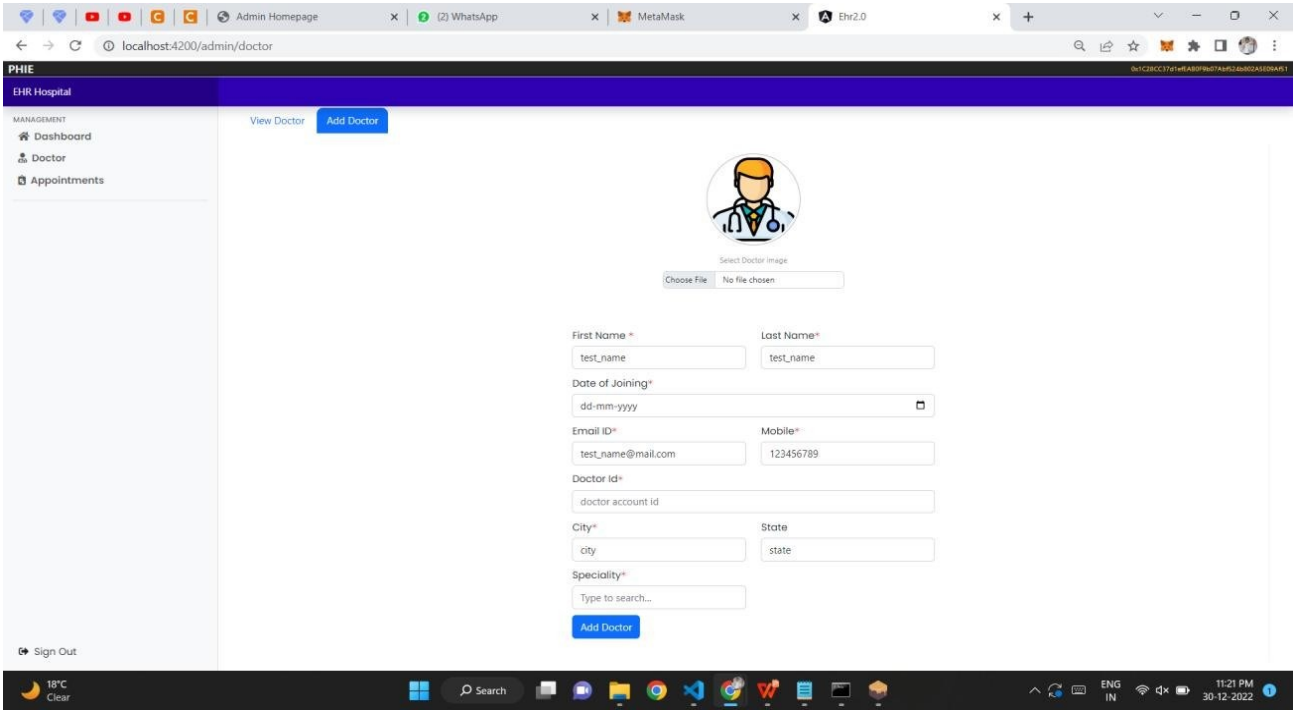


Fig: Doctor Registration

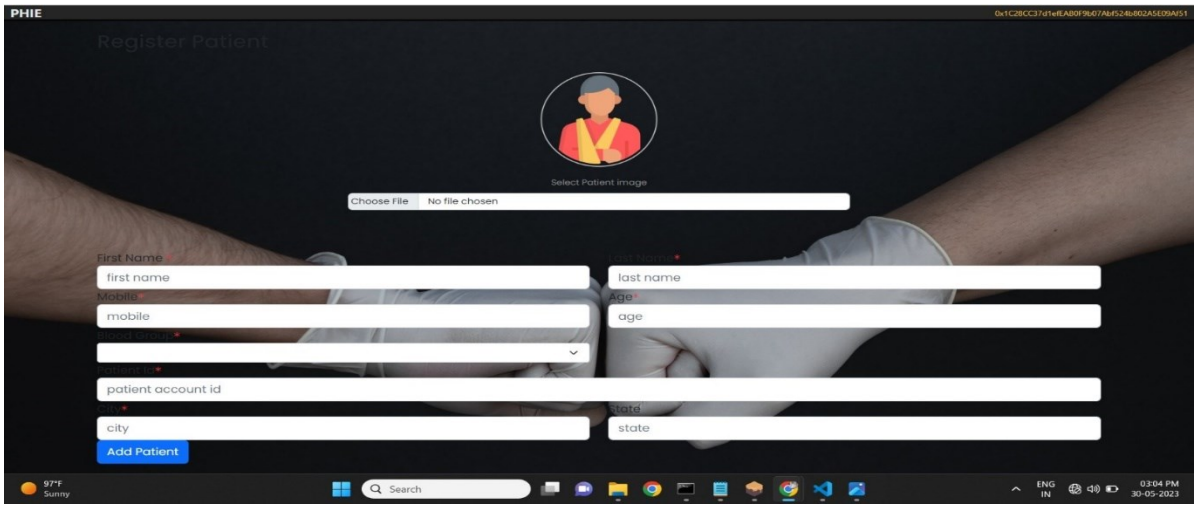


Fig: Patient Registration

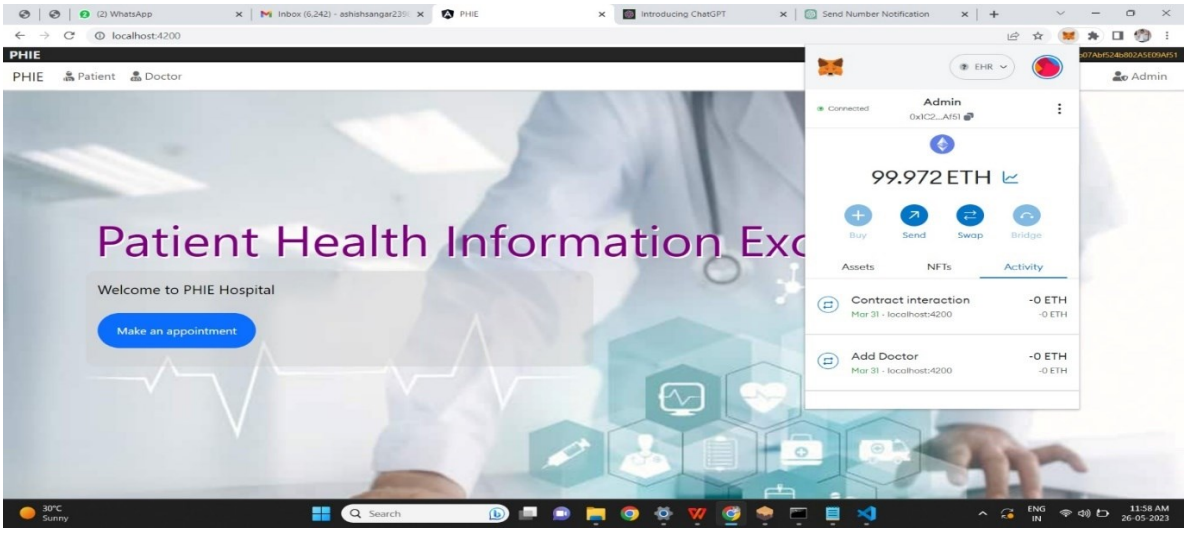


Fig. MetaMask Extension

This is MetaMask browser extension. In this various account we can connect to the blockchain network like admin, Doctor, Patient.

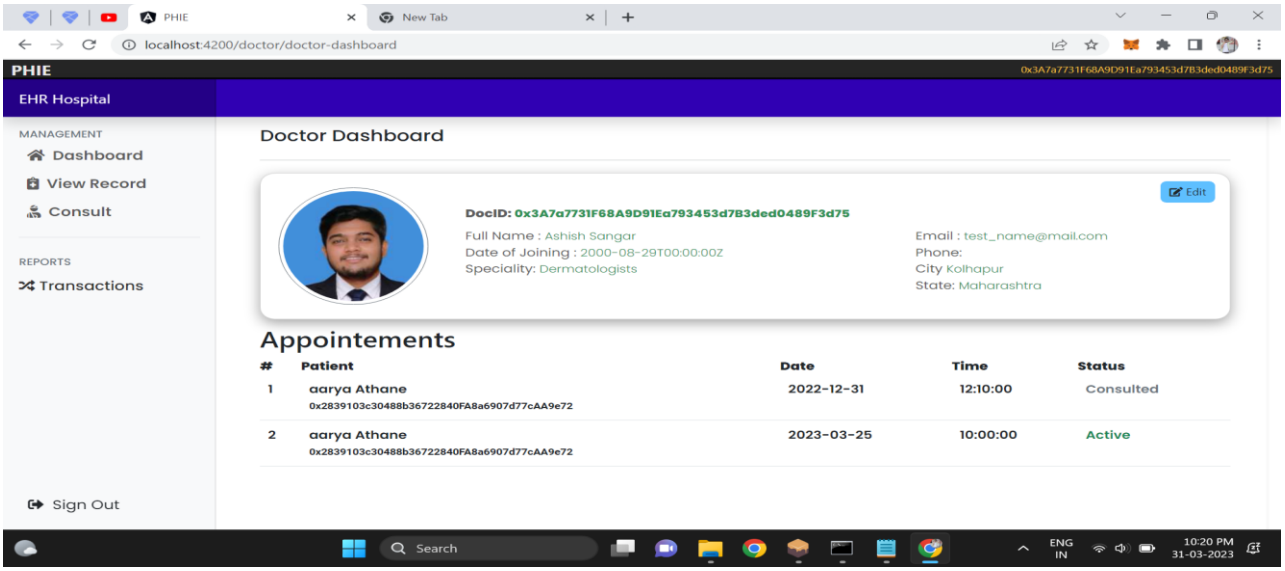


Fig. Doctor Dashboard

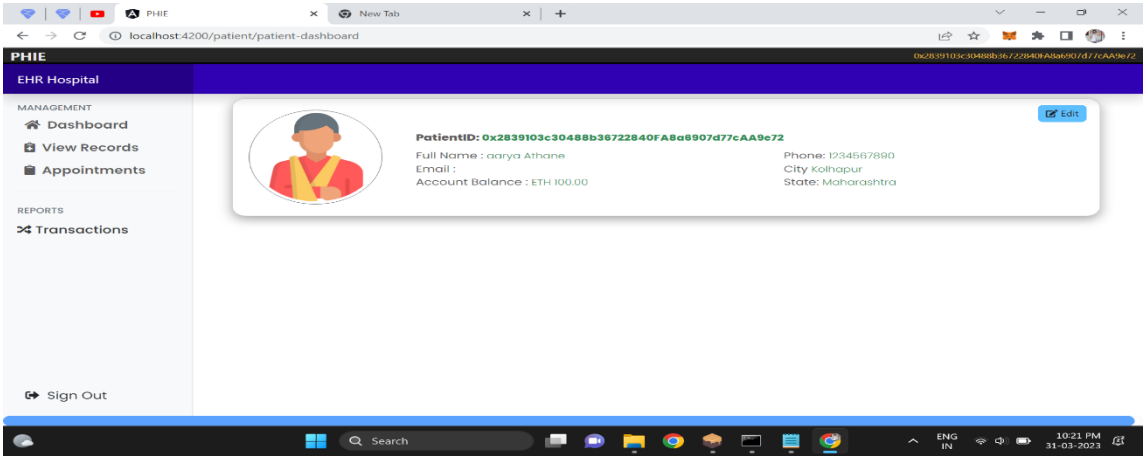


Fig. patient Dashboard

In this patient personal information is available like name, age, gender, blood group etc. Here is one button edit that gives permission to patient to edit personal information of him.

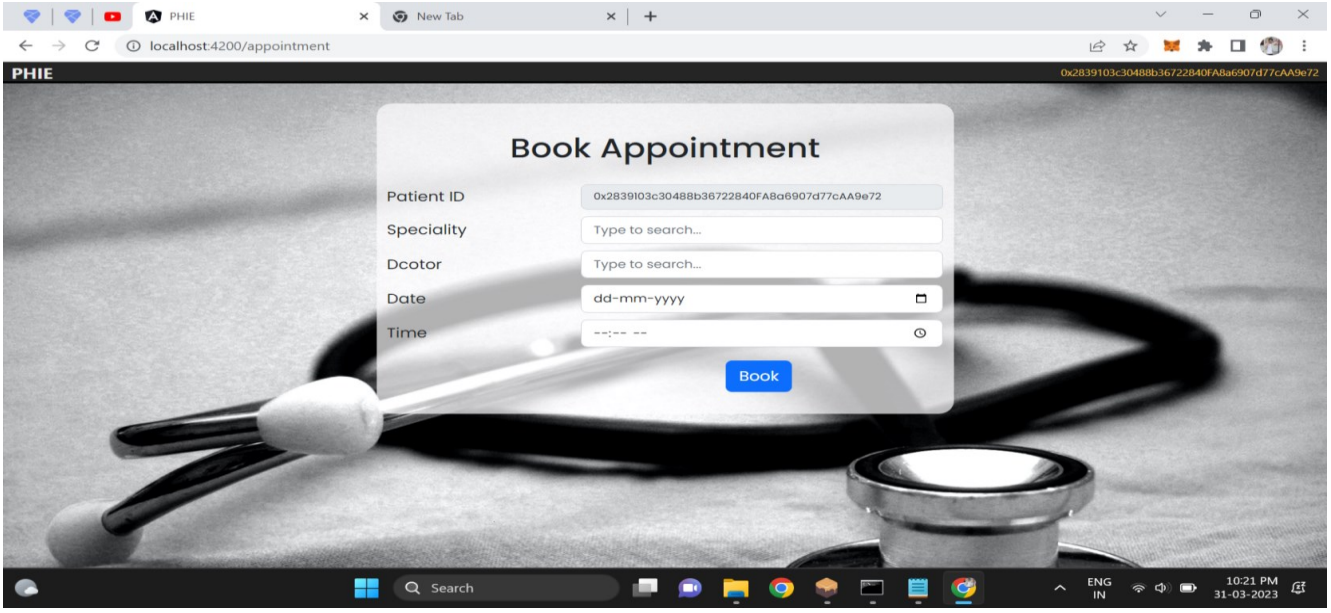


Fig. Appointment

Here patient should be felling the appointment form using patients public key as patient id and take appointment from the system.

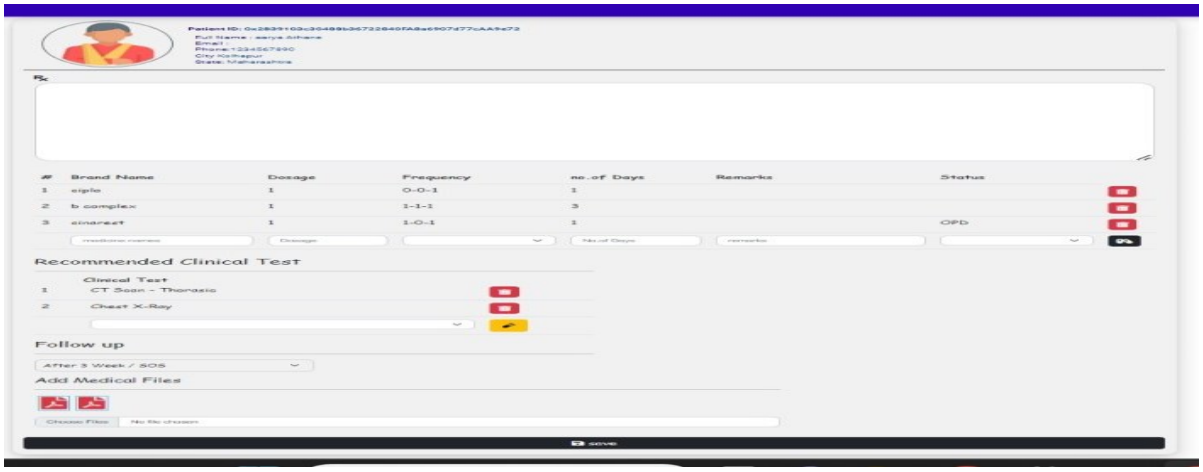


Fig. create Records

It would create patient's medical records in the DApp. It contains the fields of ID, name, blood group, and IPFS hash. The patient's basic medical records are stored along with the IPFS hash that contains the file uploaded containing the lab results or other medical records of patient.

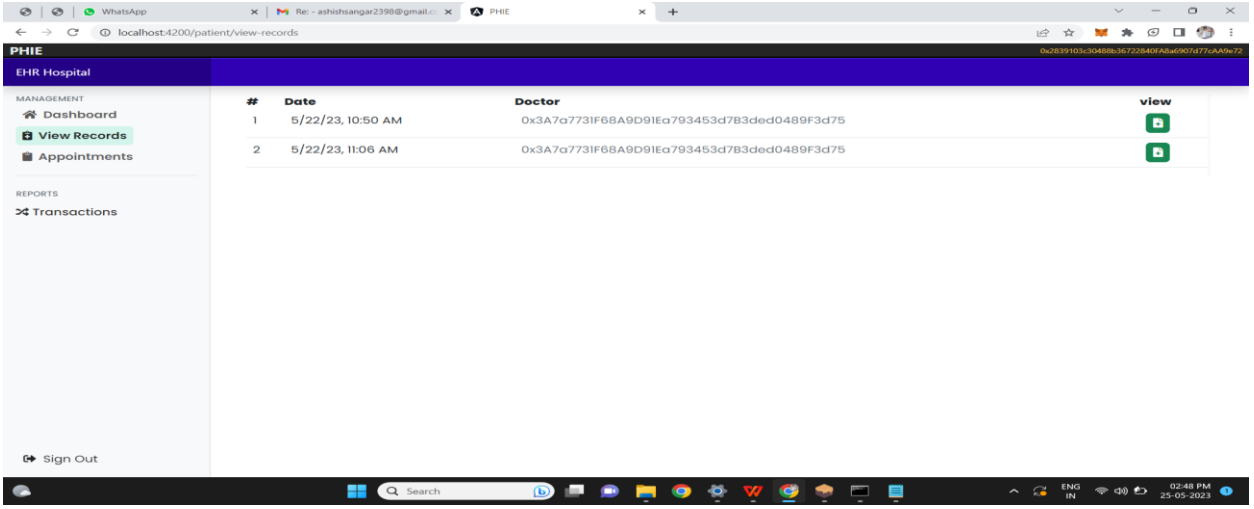


Fig. View Records from Patient Dashboard

Here patients can see his medical records which is created by doctor and also see the doctor who give treatment to patient.

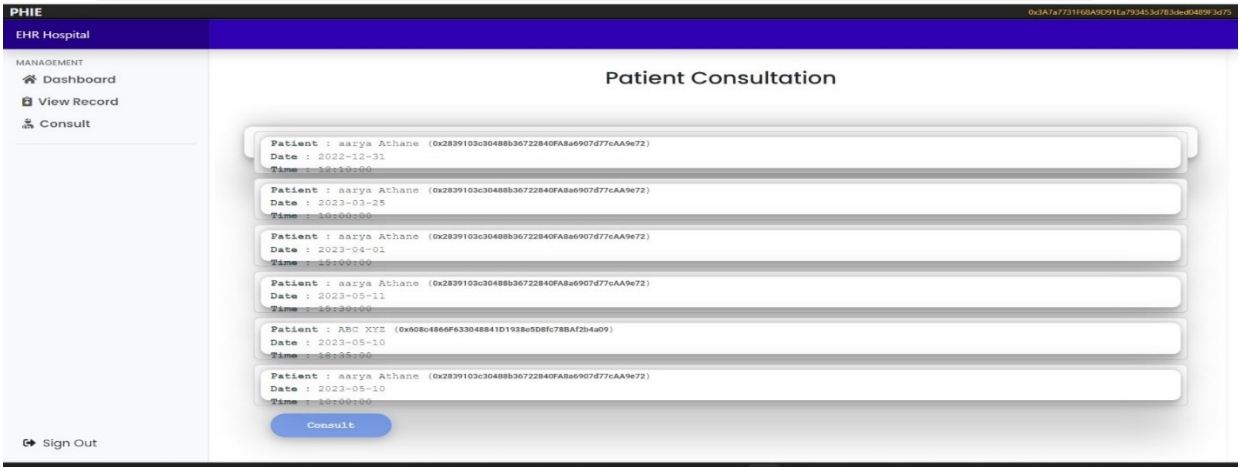


Fig. Consult Patient

From here doctor can see the appointment taken by the patient and consult the patient.

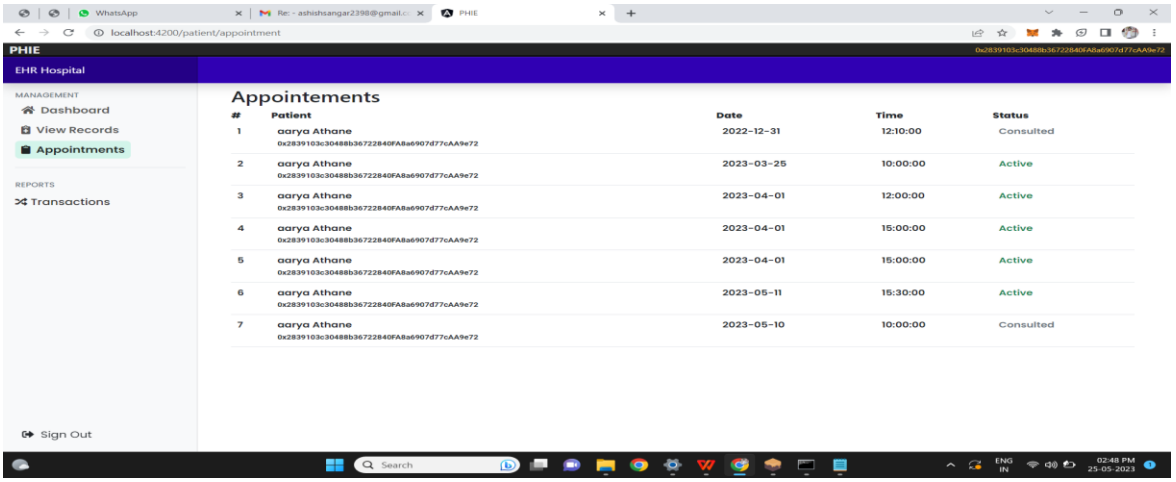


Fig . View appointment for patient

IN this Patient can see the which appointments are consulted and which not yet consulted. The Not consulting appointment shown in active status.

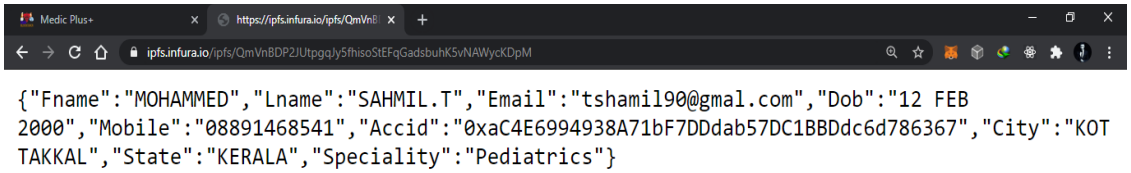


Fig .json file

The Patient Medical Record are converted to JSON object and Send to IPFS and IPFS return. The Hash which then we can store in Blockchain, and we can use it to retrieve the record.

TESTING:

TESTING MECHANISM

Unit Testing

The goal of unit testing is to isolate each part of the program and show that the individual parts are correct. Unit testing is essential for verification of the code produced during the coding phase, and hence the goal is to test the internal logic of the modules.

Test Cases : -

Problem summary: Improve medical record exchange process and access of health records by blockchain.

Test Case 1:-

Name: Admin Registration

Input: click admin icon on the home screen

Logical process:

1.Write password of admin account

2. admin dashboard is displayed

Test Case 2:

Name : Patient registration

Logical process

1. Click on the patient icon

2 Display the Patient registration form

3 Use unique id as patients public key

4.fill all the mentioned information

5 click on register button

Test Case 3:-

Name: Doctor Registration

Input: The (doctor) clicks on the doctor icon on the home screen

Logical process:

1. Click on the doctor icon

2. Connect doctor account to MetaMask

3. Displays the registration form

Test Case 4:-

Name: consult patient

Input: The user doctor clicks on the consult icon on the doctor's sidebar screen

Logical process:

1 Clicks on the consult icon.

2.put patient's public key

3.view update and delete record

4.Saves the medical document in database for future use

Test Case 5:

Name: upload

Input: The user(doctor) clicks on the view records on the dashboard

Logical process:

1. Clicks on the view records icon

2.Displays the previous information if that patient

3. Doctor can add new prescription

Test Case 6:

Name: Create Report

Input: The user(doctor) Click on the create report icon on the screen

Logical Process:

1. Click on the Create Report icon

2. Click here to select image Doctor can select multiple images.

3. Give the file name and save it

4. Click on select images

5. Select multiple images.

6. Check previous and next option also available

CONCLUSION & FURTHER WORK

In this project we discussed how blockchain technology can be useful for healthcare sector and how can it be used for exchange electronic health records. Despite the advancement in healthcare sector and technological innovation in EHR systems they still faced some issues that were addressed by this novel technology, i.e., blockchain. Our proposed framework is a combination of secure record storage along with the granular access rules for those records. It creates such a system that is easier for the users to use and understand. Also, the framework proposes measures to ensure the system tackles the problem of data storage as it utilizes the off-chain storage mechanism of IPFS. And the role-based access also benefits the system as the medical records are only available to the trustedand related individuals. This also solves the problem of information asymmetry of EHR system.

For example, developing machine learning algorithms that can look at a radiology image orCT scan and make diagnosis predictions are time consuming to develop, but once developedare easy to execute. By leveraging blockchain technology, one can envision a world in which a service provider publishes FHIR Diagnostic Reports of radiology images to the blockchain.

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